

Choosing dashboard tools

DASHBOARD DESIGN CONCEPTS



Olga Scrivner, PhD
President, Scrivner Solutions Inc

How to choose a dashboard software?

Dashboard software is an information management tool for tracking and displaying data

Traditional
dashboards

BI/IT team performs the data analysis
work for business users

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Software architecture	Drag-and-drop applications, dashboard web frameworks

Drag-and-drop dashboard applications

- No technical background required
- Limited design customization



Introduction to Power BI

Learn how to navigate and use Power BI, a popular data analysis and visualization platform, to create compelling reports.



Data Analysis in Power BI

Discover Exploratory Data Analysis (EDA) on Power BI and learn how to extract the kind of insights that lead to powerful reports.



Data Visualization in Power BI

Take your reports and dashboards to the next level with Power BI, a data visualization tool.



Data Modeling in Power BI

Find out how the Power BI platform can power up your data modeling.





Introduction to Tableau

Learn how to use Tableau, a popular data analytics and business intelligence (BI) program, to manage data and create visualizations.



Analyzing Data in Tableau

Unlock the power of advanced analytics and data visualizations in Tableau.



Creating Dashboards in Tableau

Discover how to create impactful dashboards in Tableau and boost business performance.



Case Study: Analyzing Customer Churn in Tableau

The fictional company Databel needs your help! Use your Tableau skills to analyze customer churn data and discover potential solutions.



Dashboard frameworks

- Technical background required



Building Web Applications with Shiny in R

Shiny is an R package that makes it easy to build interactive web apps directly in R, allowing your team to explore y...



Case Studies: Building Web Applications with Shiny in R

Practice your Shiny skills while building some fun Shiny apps for real-life scenarios!



Building Dashboards with shinydashboard

In this course you'll learn to build dashboards using the shinydashboard package.



Building Dashboards with flexdashboard

In this course you'll learn how to create static and interactive dashboards using flexdashboard and shiny.



- Full control of dashboard design



Building Dashboards with Dash and Plotly

Learn how to build interactive and insight-rich dashboards with Dash and Plotly.



Introduction to Data Visualization with Plotly in Python

Create interactive data visualizations in Python using Plotly.

Python Tutorial: Streamlit

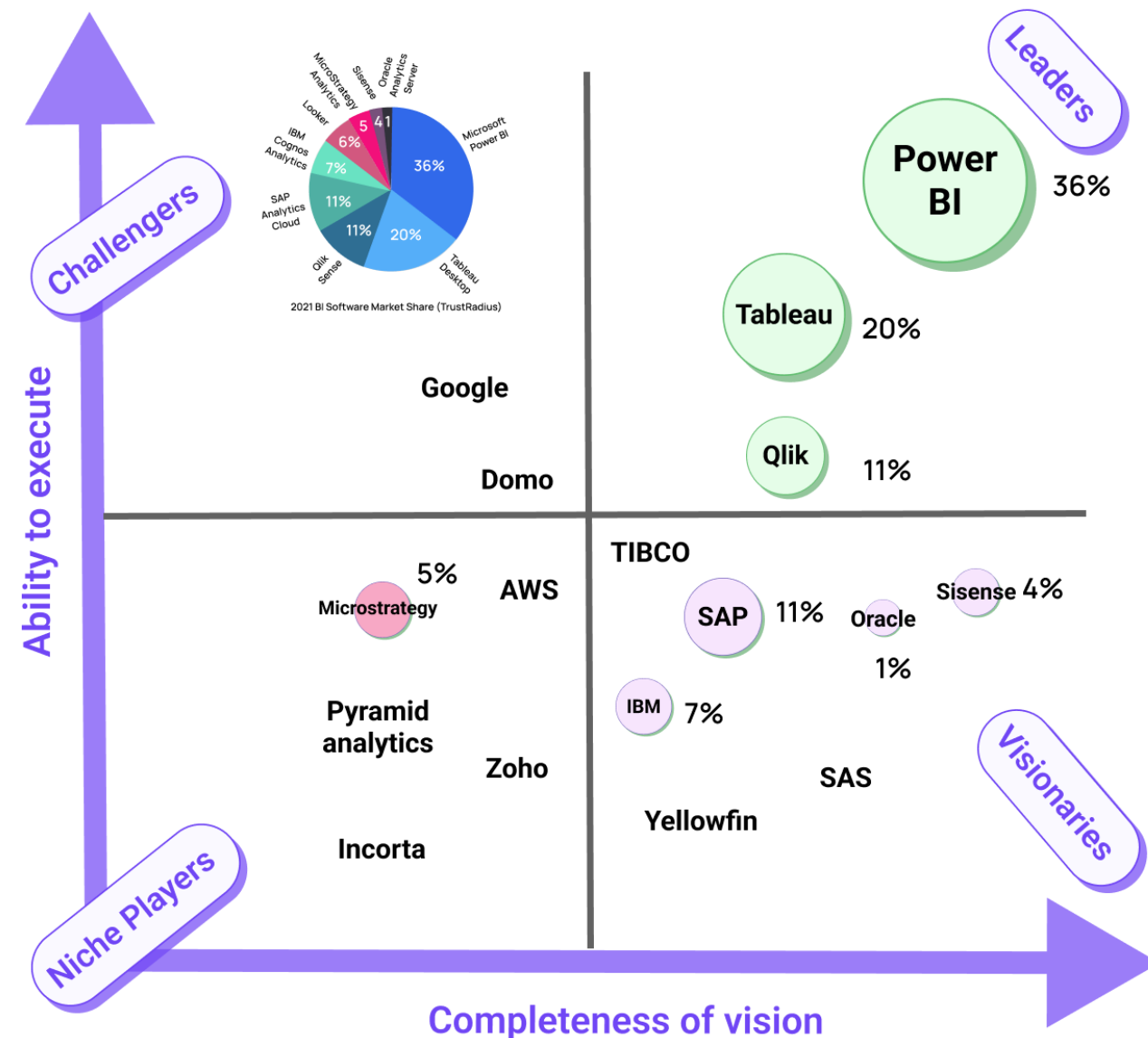
This tutorial about Streamlit is meant to help Data scientists or machine learning engineers who are not web developers and they're not interested in spending weeks learning to use these frameworks to build web apps.

Python's Dash Package



Top dashboard software: market share and interest

Marketshare proportion (TrustRadius, 2021)



Microsoft Power BI: desktop (windows), cloud service (Azure), integration with Microsoft products

Tableau: Tableau Server, Tableau Desktop, easy sharing on Tableau Public gallery

Qlik Sense: desktop, cloud-based, integration with cloud services (Azure, AWS, Google)

SAP Analytics Cloud: a cloud software for business intelligence, planning, integration with other cloud services

¹ Gartner Magic Quadrant. (2022). BI and Analytics

Let's practice!
DASHBOARD DESIGN CONCEPTS

Dashboard checklist

DASHBOARD DESIGN CONCEPTS



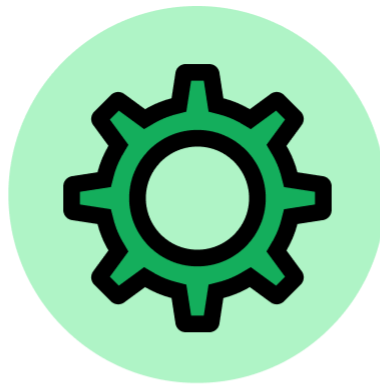
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Dashboard design constructs



Dashboard

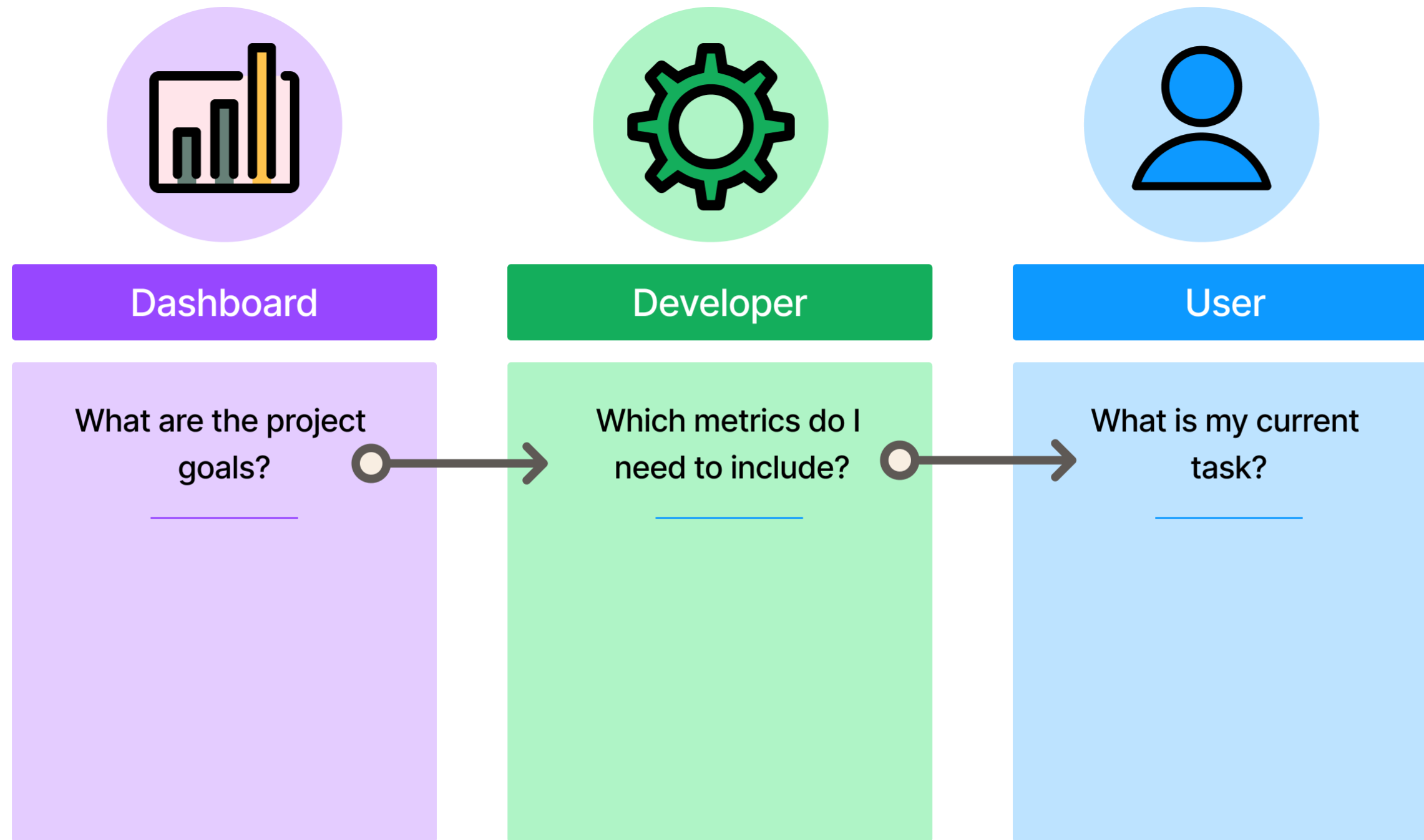


Developer

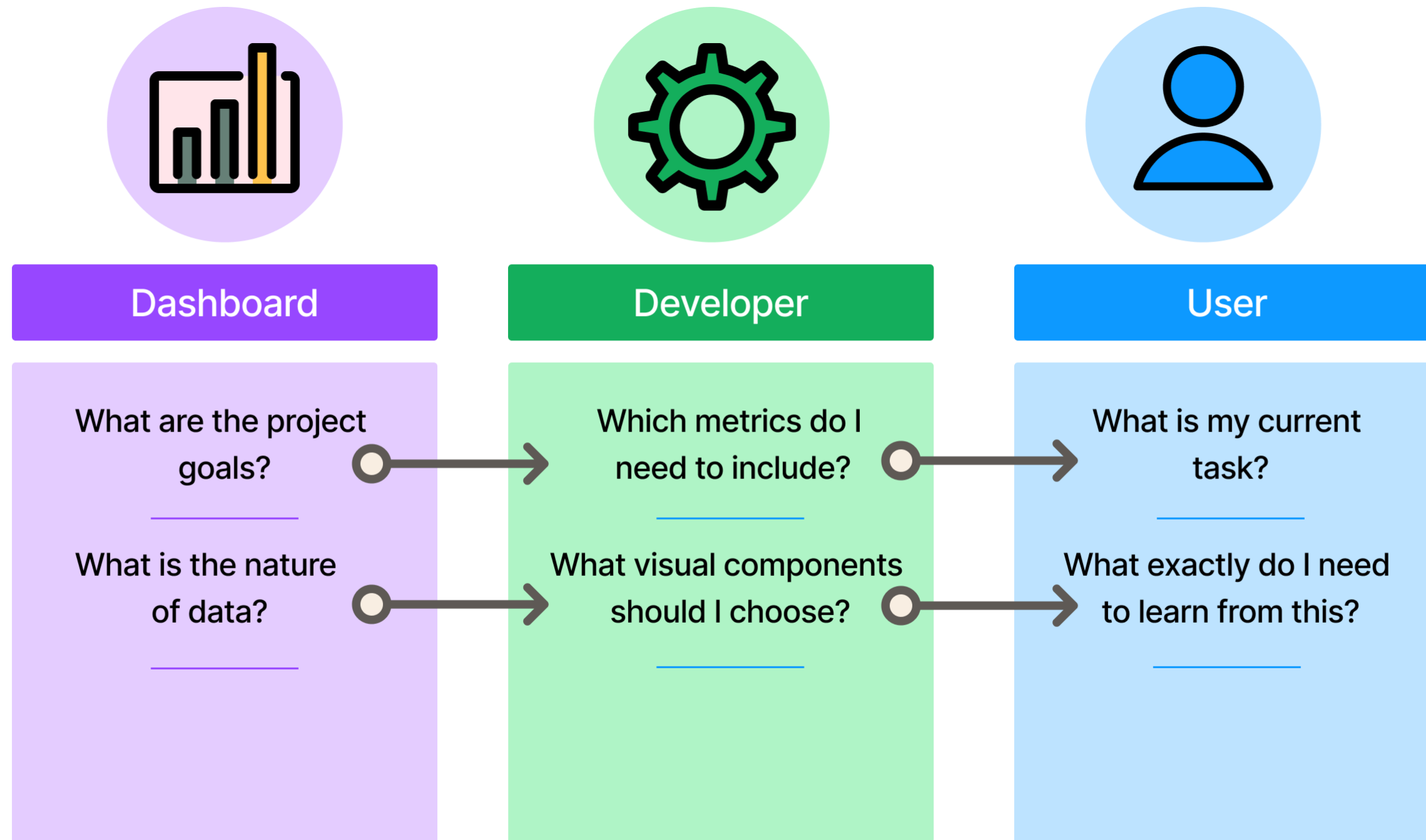


User

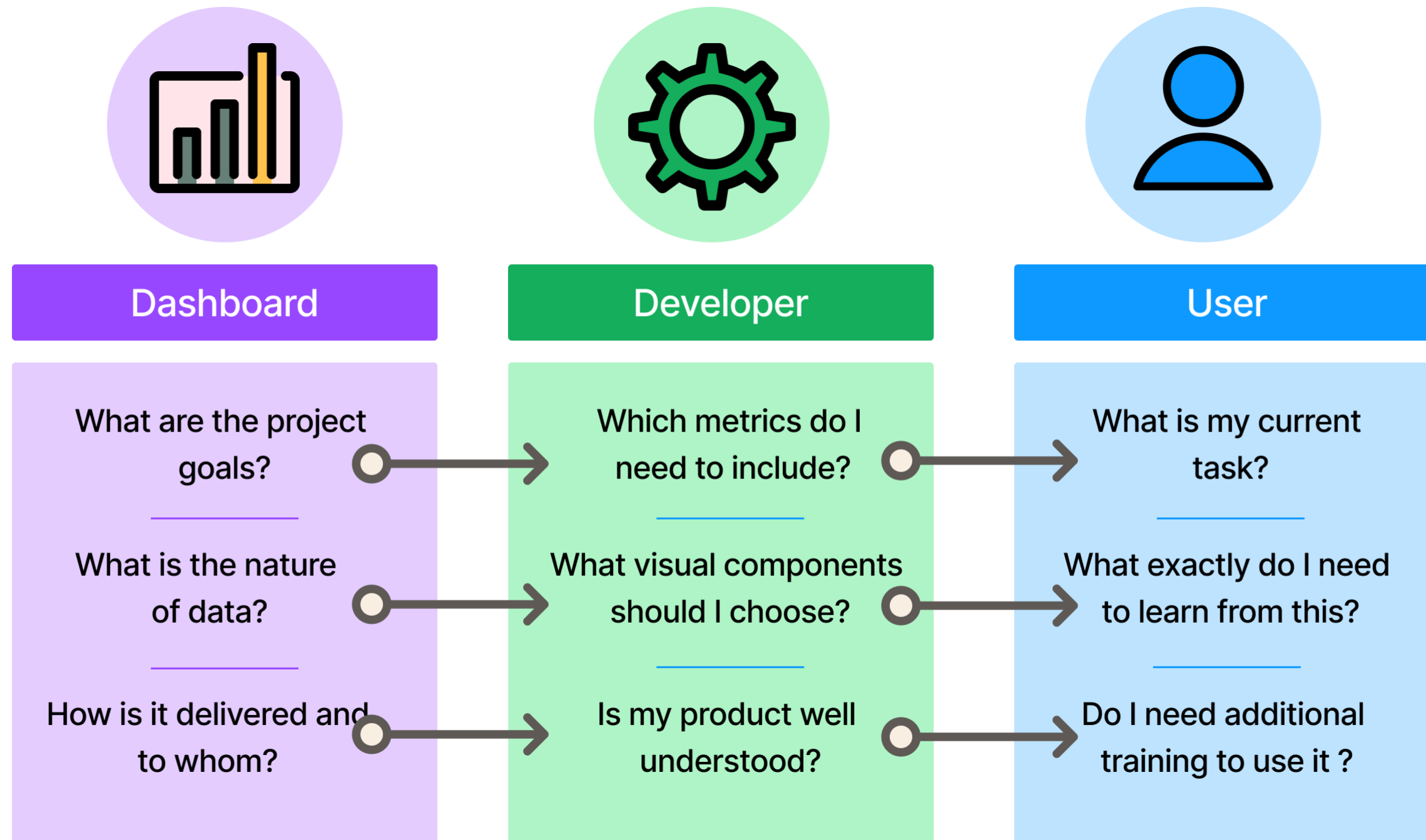
Dashboard design units



Dashboard design units



Dashboard design units



Dashboard design assessment checklist



Practical test
Does it serve its purpose?

¹ Sisense. (2022). Assessing the Quality of Your Dashboard.

Dashboard design assessment checklist



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Data relevance test
Does it display relevant information?

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Is the information displayed correctly?

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Is it intuitive for use?

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Dashboard design assessment checklist



Practical test
Does it serve its purpose?



Data relevance test
Does it display relevant information?



Visual correctness test
Is the information displayed correctly?



Intuitiveness test
Is it intuitive for use?



Adoption test
Is it adopted by the users?

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Know your dashboard type



Strategic: displays high-level metrics and a small number of plots

Tactical: displays many metrics over a large amount of data related to a single area of business

Analytical: displays many metrics related to enterprise-wide historical data for in-depth analysis

Operational: displays granular details related to a particular business task

Know your audience

Internal stakeholder

Use company's terms, acronyms,
less context is needed

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External stakeholder

Use less acronyms, add more context

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Quick-glancing

Highlight insights, use fewer charts

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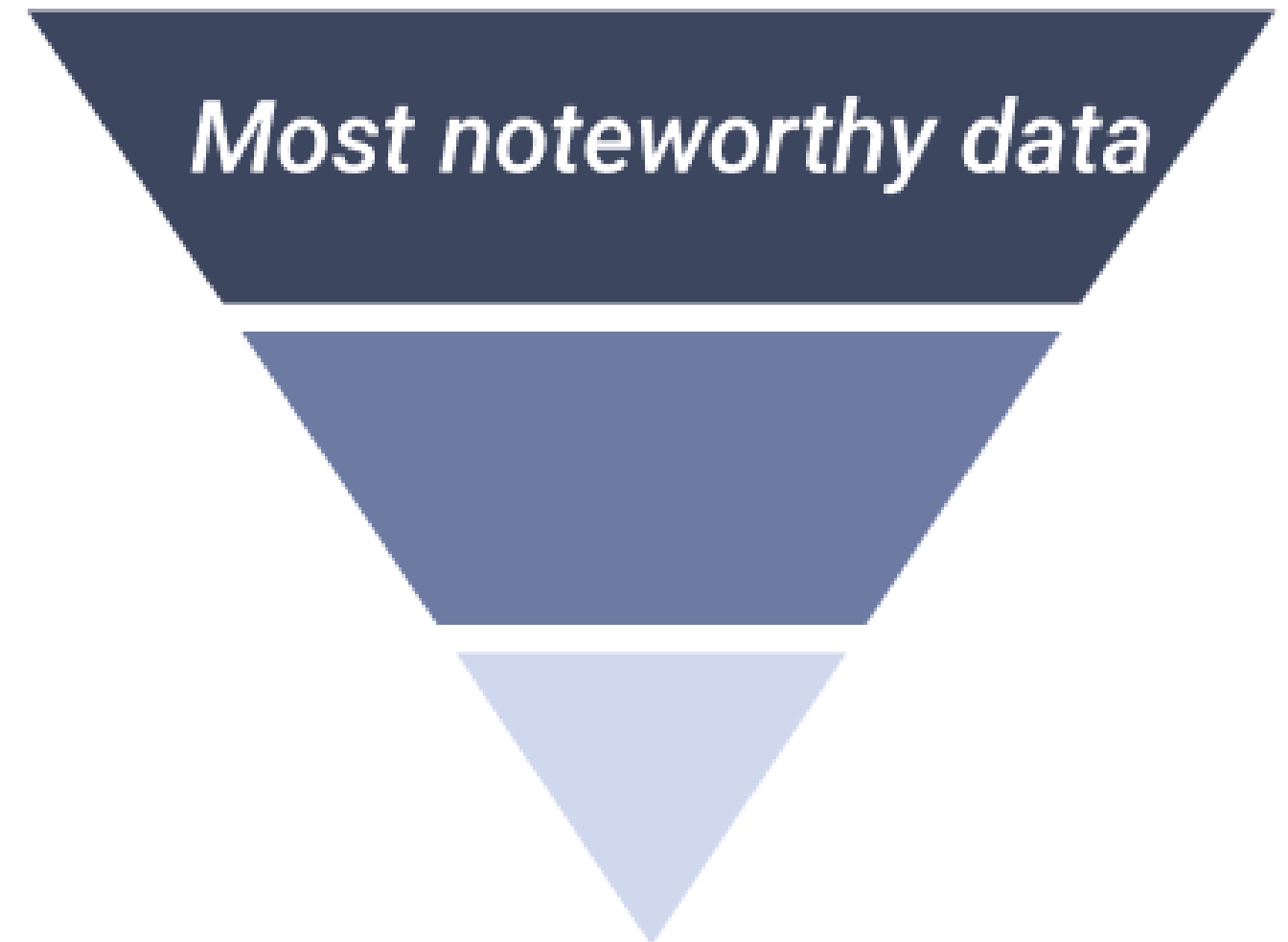
Quick-glancing

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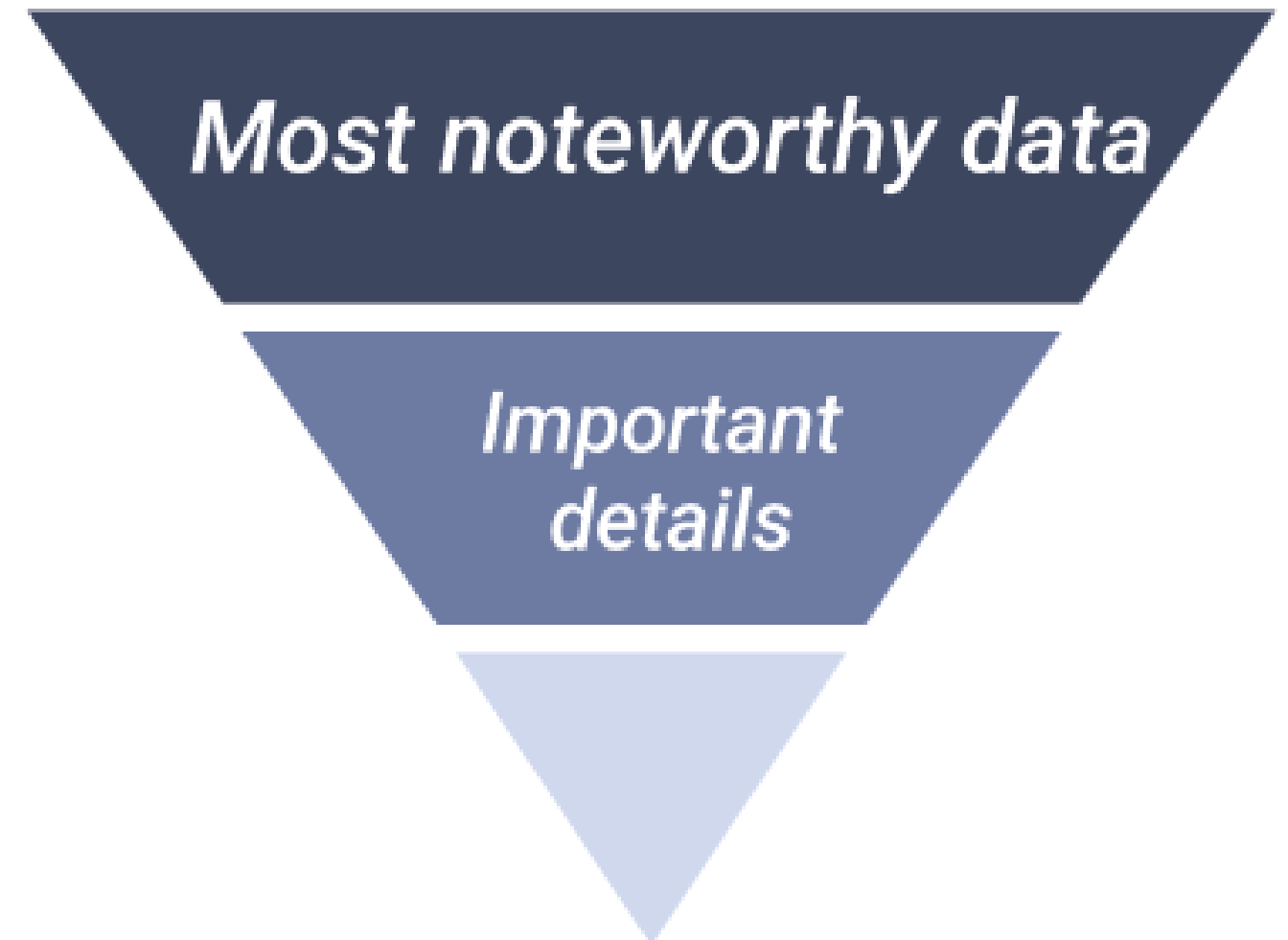
Long-lingering

Use more charts and drill-down data

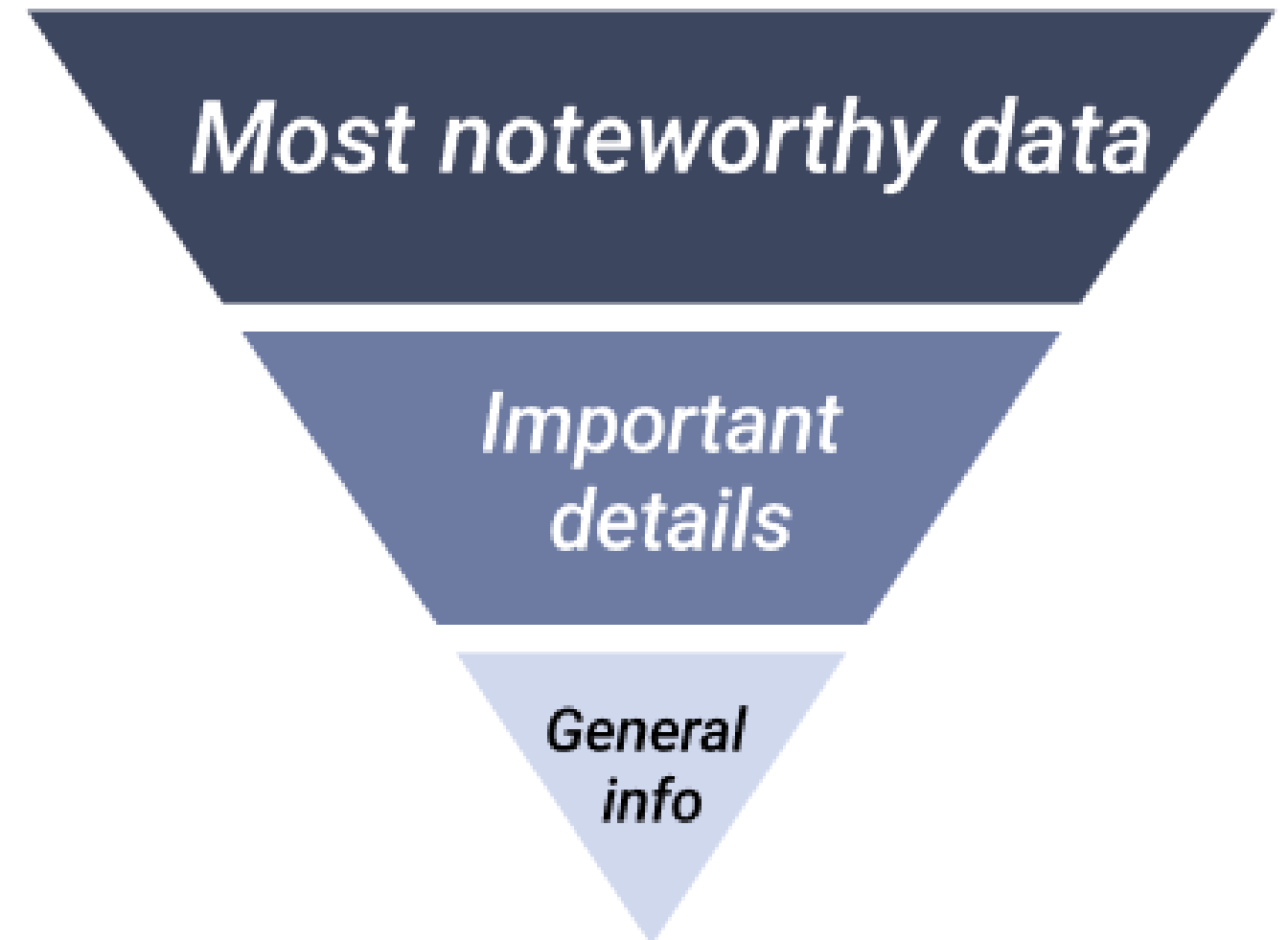
Design your dashboard with a narrative in mind



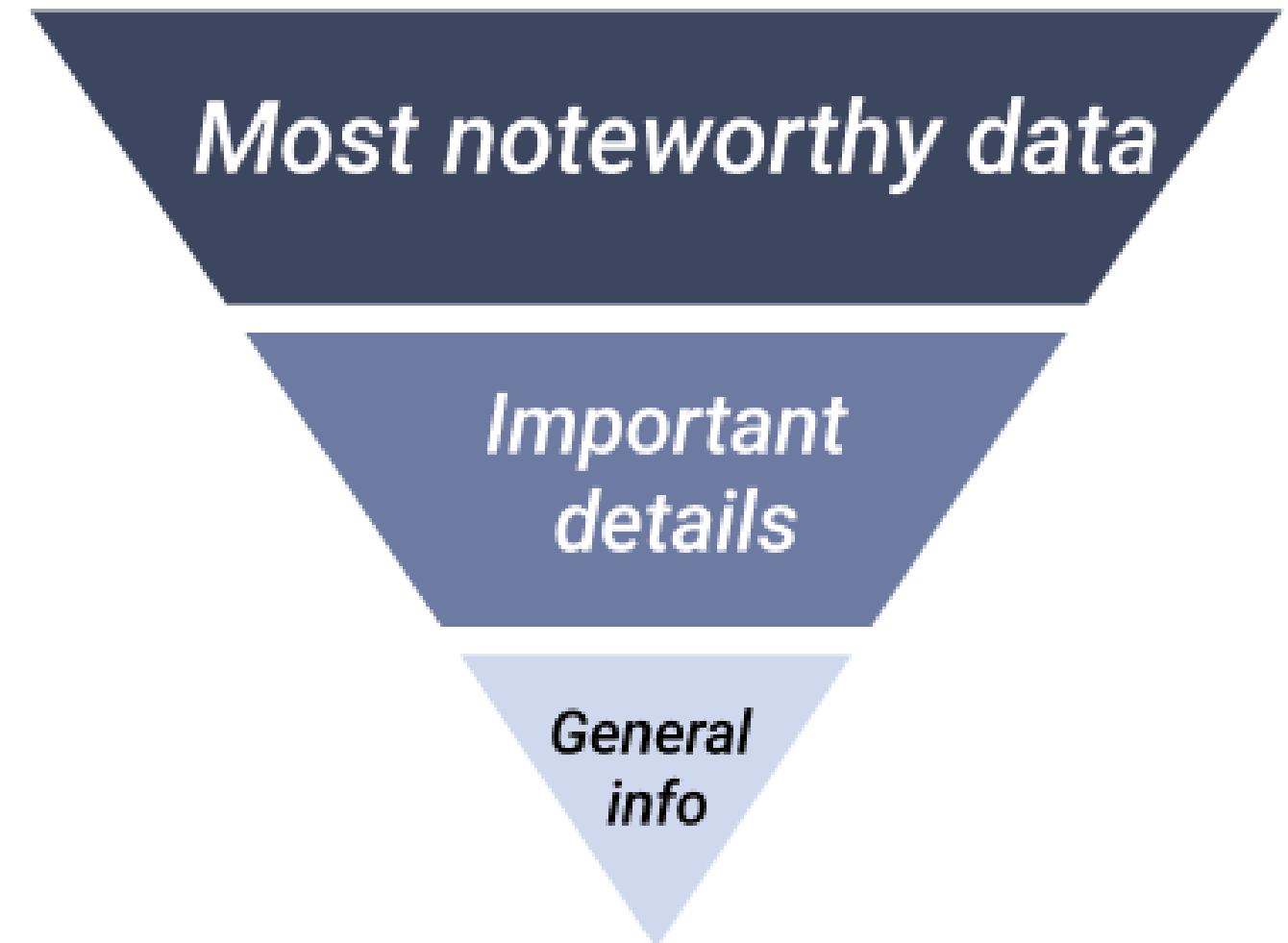
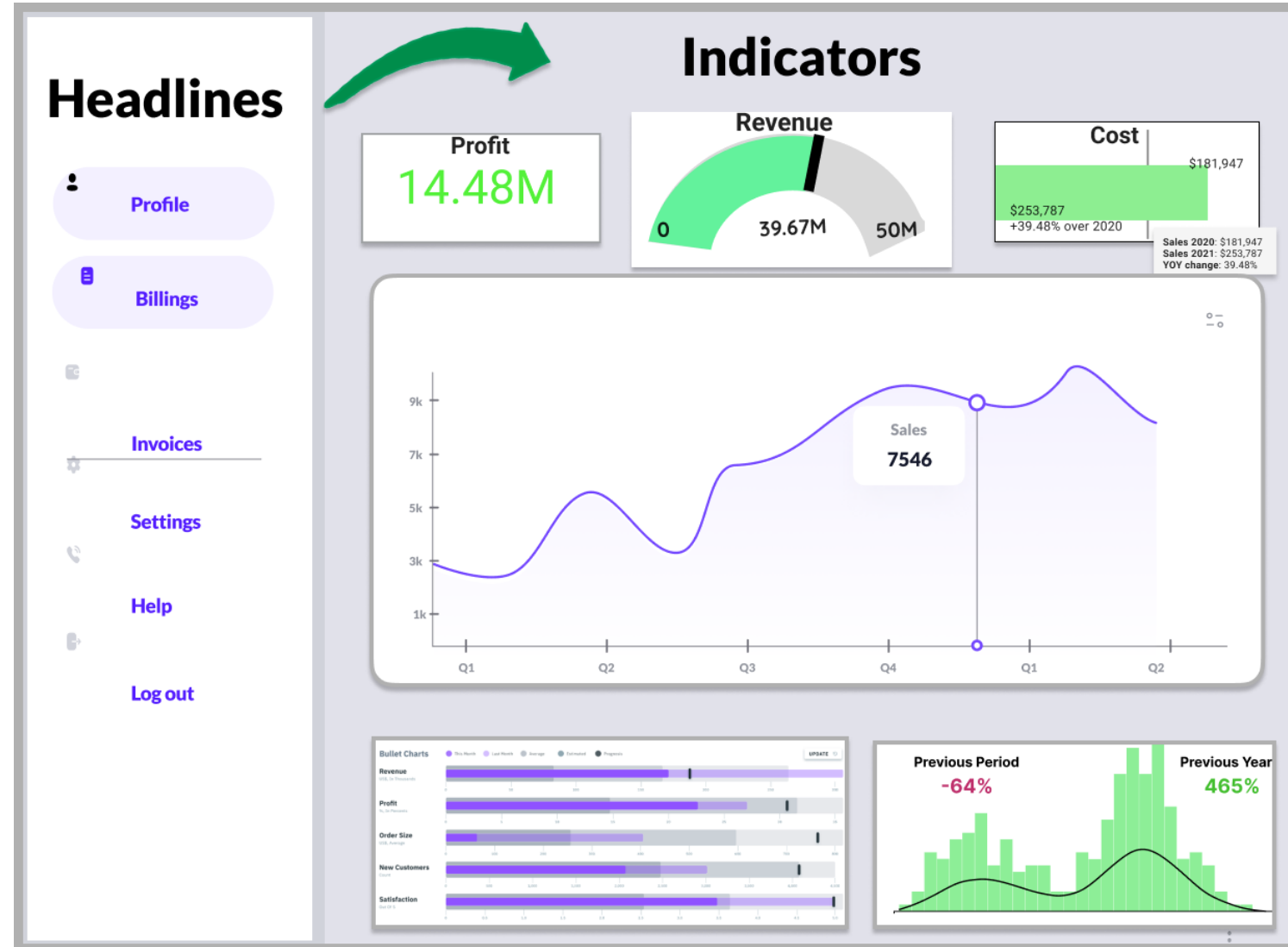
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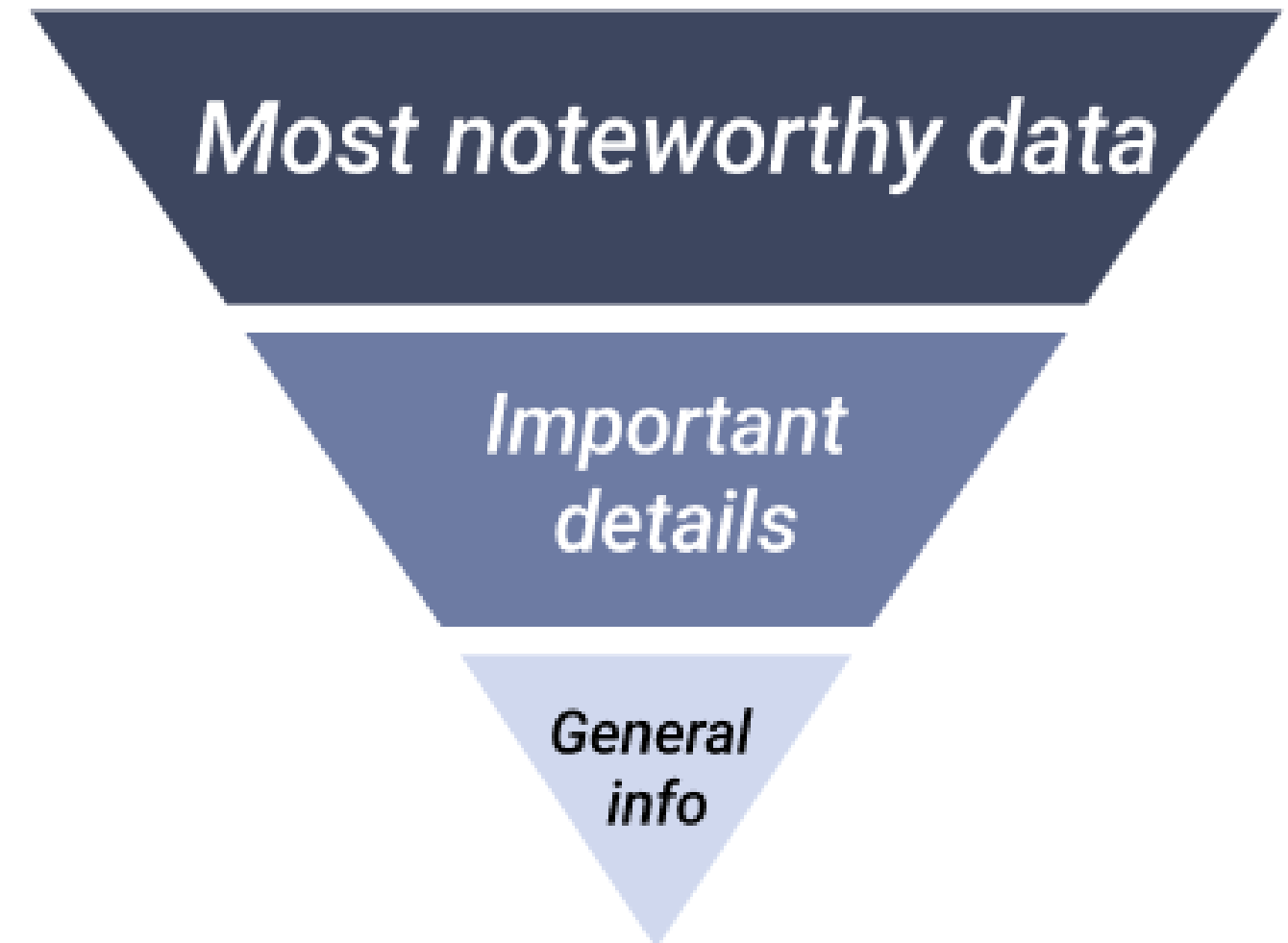
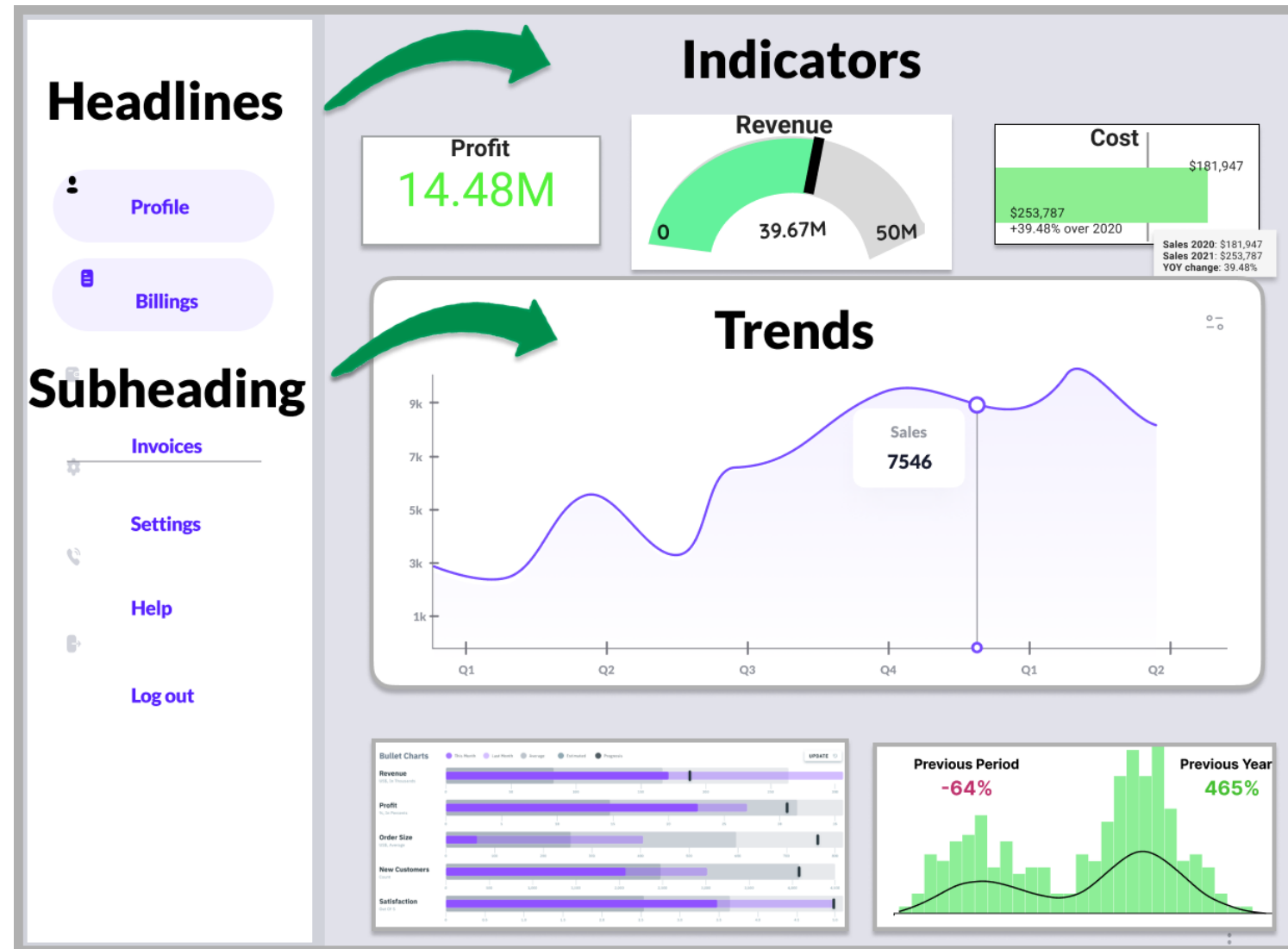
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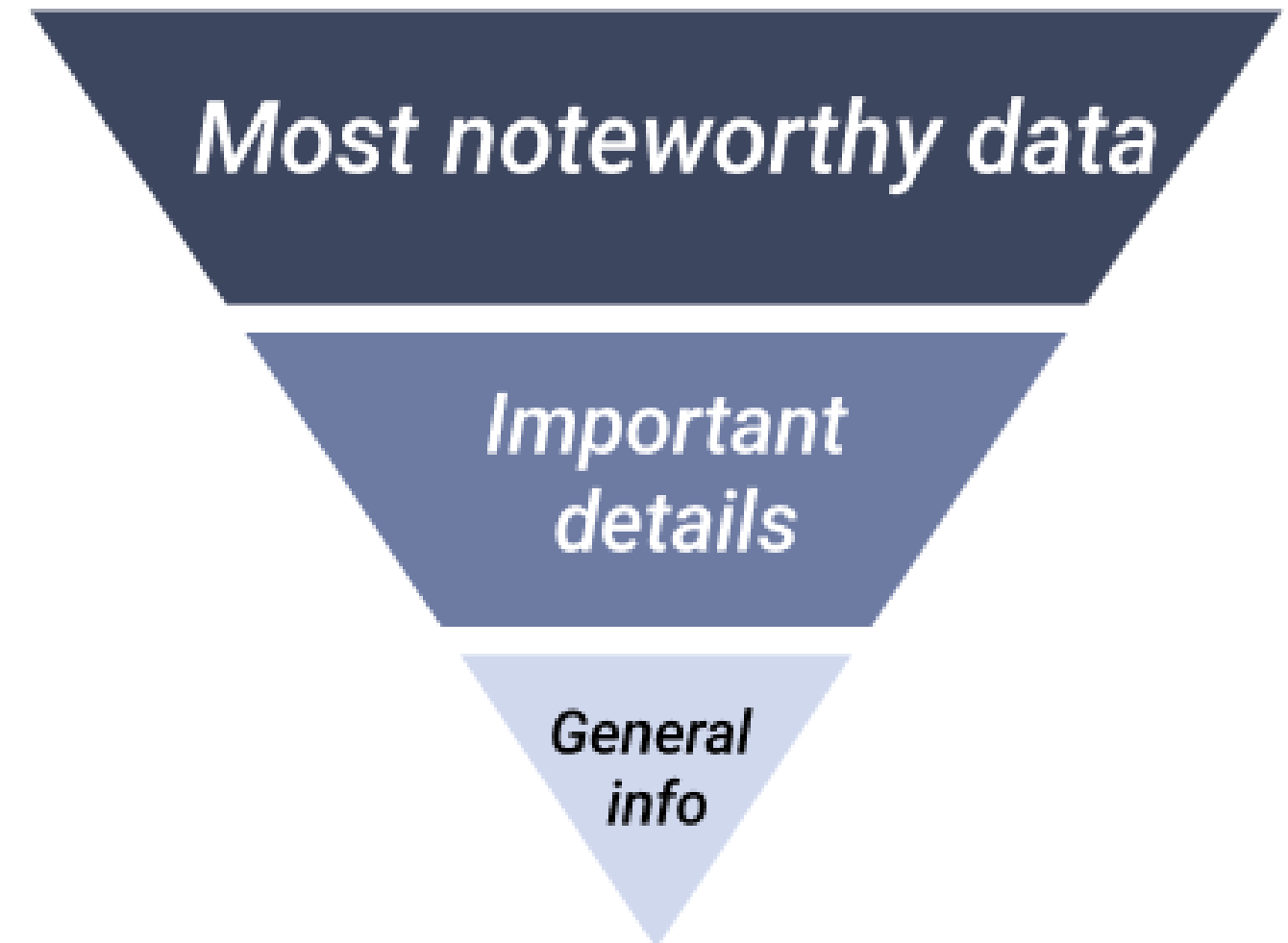
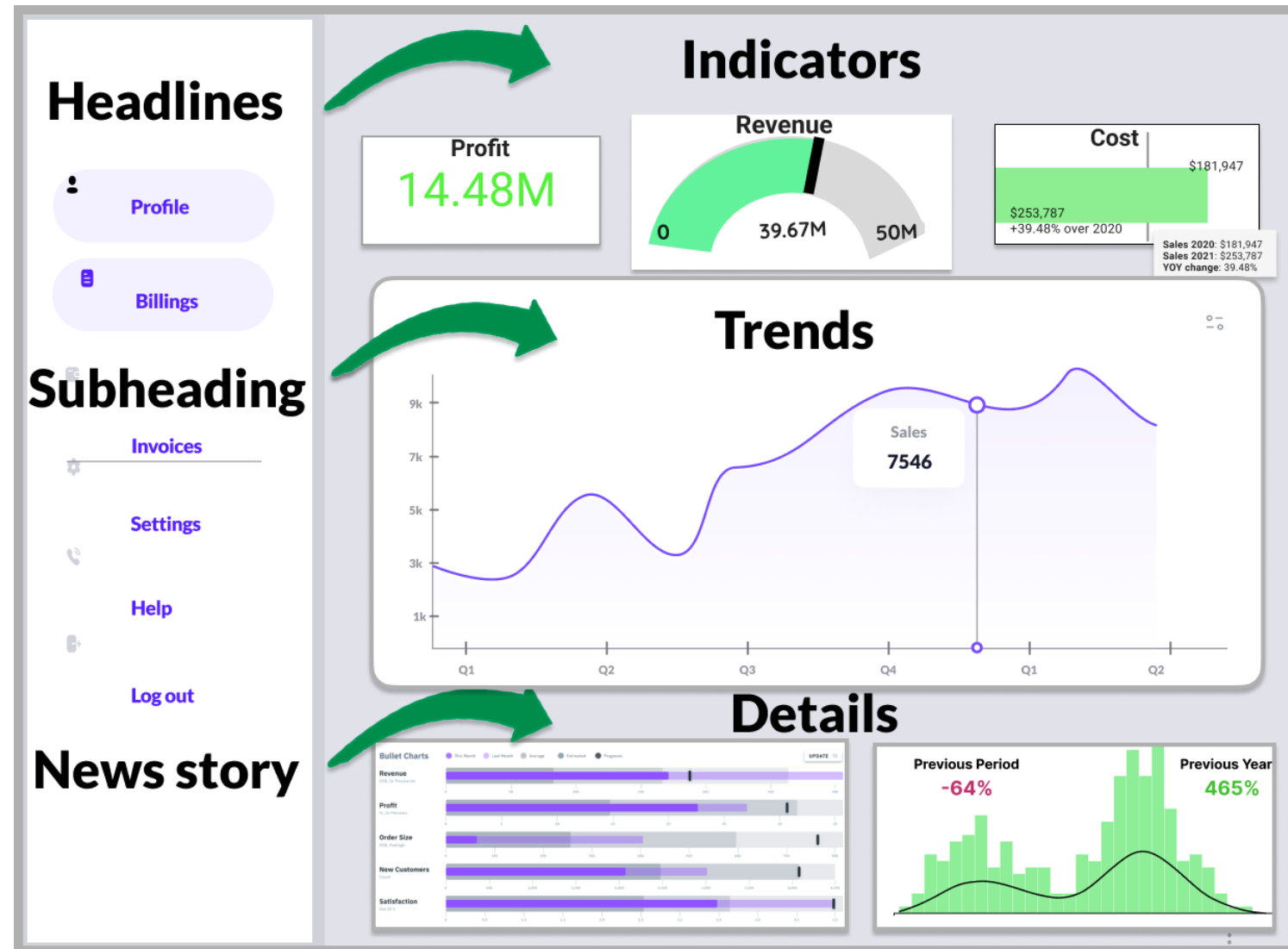
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Design your dashboard with a narrative in mind



Design your dashboard with a narrative in mind



¹ Geckoboard. (2023). Effective dashboard design

Reduce cognitive load

5

Use the 5-second rule: present the most important data in only 5 seconds

Reduce cognitive load

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Be accessible: choose easy to read fonts, color-blind palette

Reduce cognitive load

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Be consistent: repeat colors, font, and design elements

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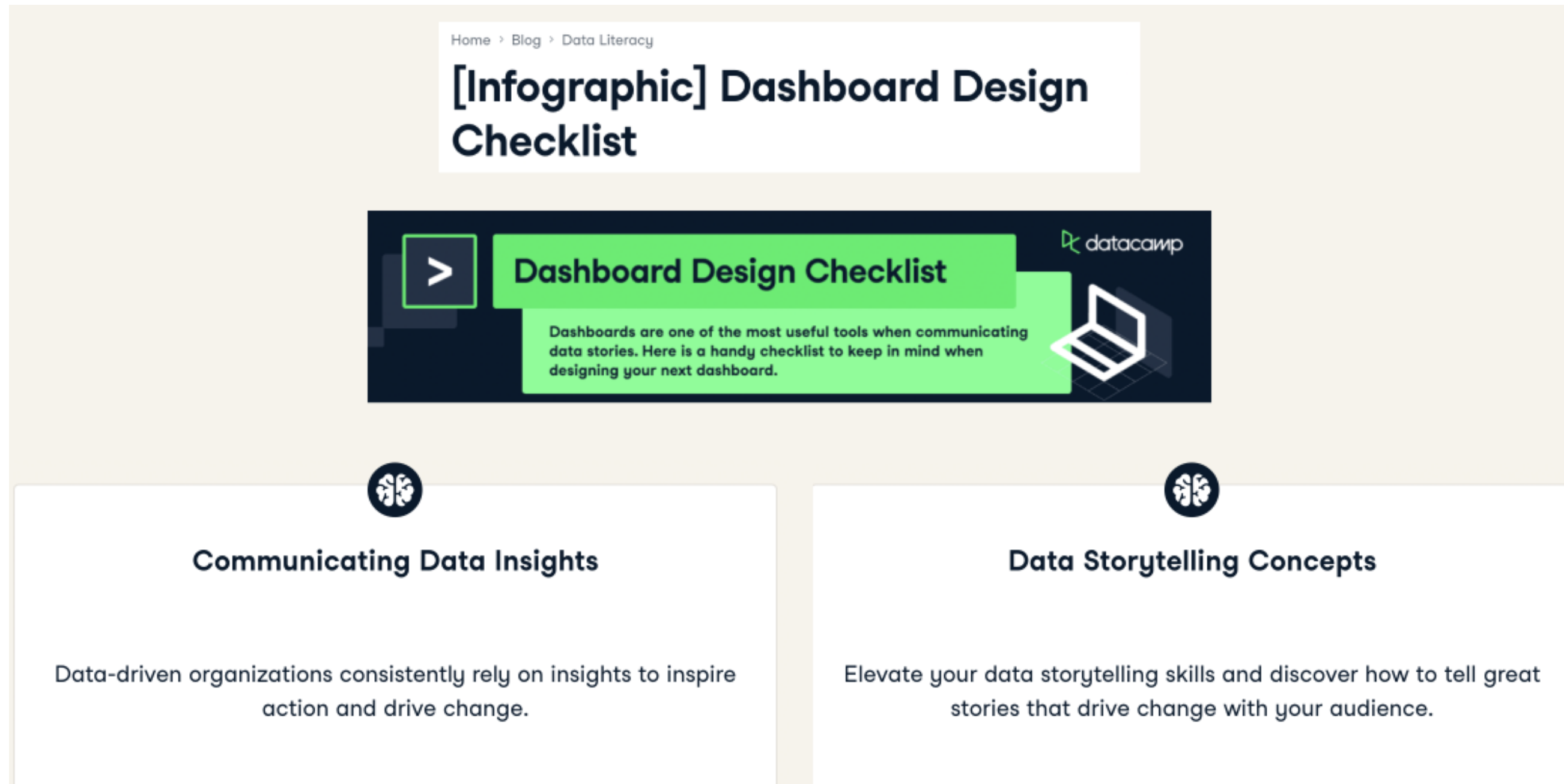


Be consistent: repeat colors, font, and design elements



Be clear: provide visual clue, context, icons

Additional thoughts



¹ <https://www.datacamp.com/blog/infographic-dashboard-design-checklist>

Let's practice!
DASHBOARD DESIGN CONCEPTS

Measuring dashboard success

DASHBOARD DESIGN CONCEPTS



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Successful dashboard benefits



Improved decision-making



Successful dashboard benefits



Improved decision-making



Increased efficiency and transparency

Successful dashboard benefits



Improved decision-making



Increased efficiency and transparency



Improved communication and collaboration

Successful dashboard benefits



Improved decision-making



Increased efficiency and transparency



Improved communication and collaboration



Enhanced customer satisfaction

Usage analytics

How many times users log in to the application?



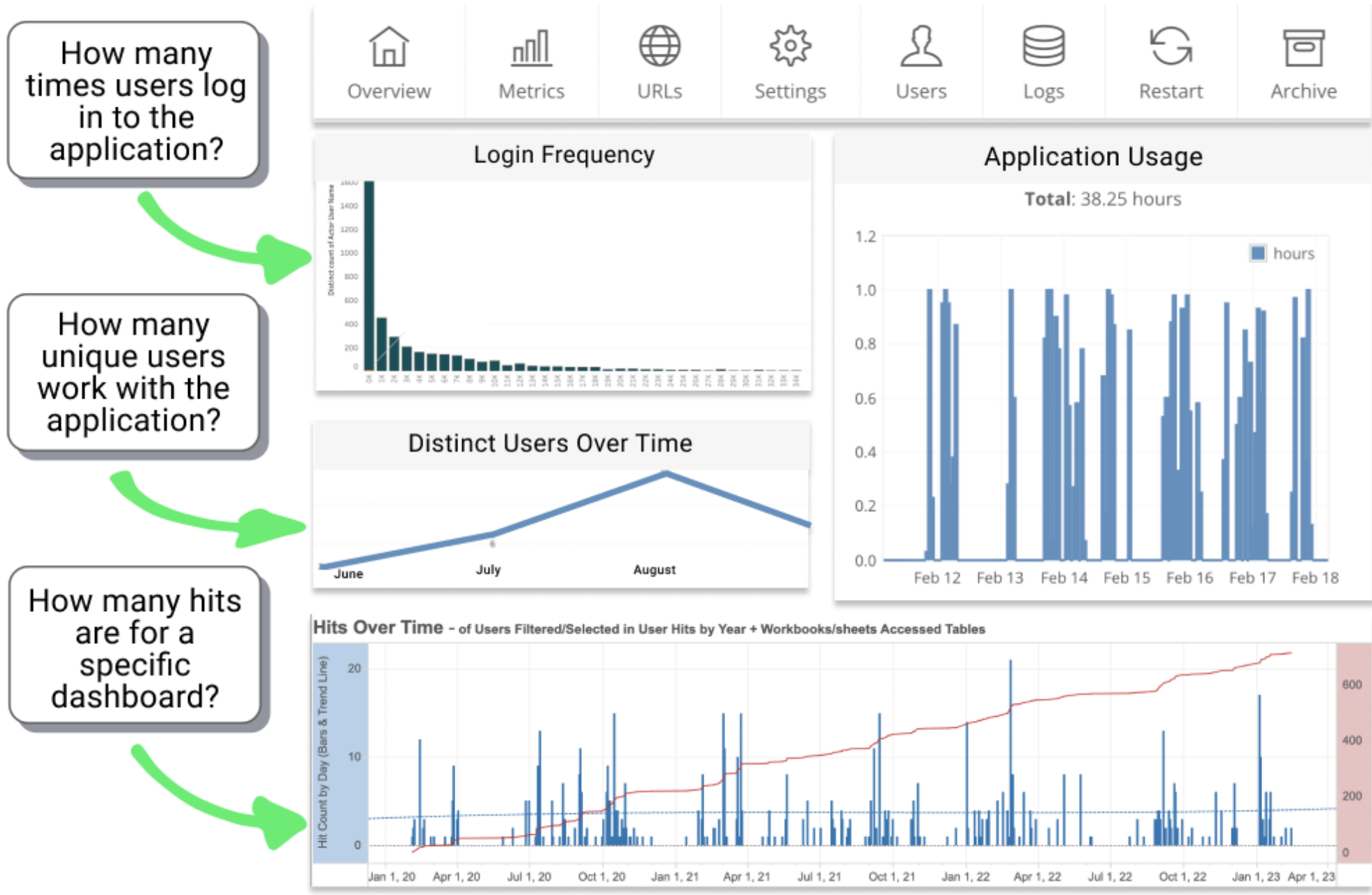
Usage analytics

How many times users log in to the application?

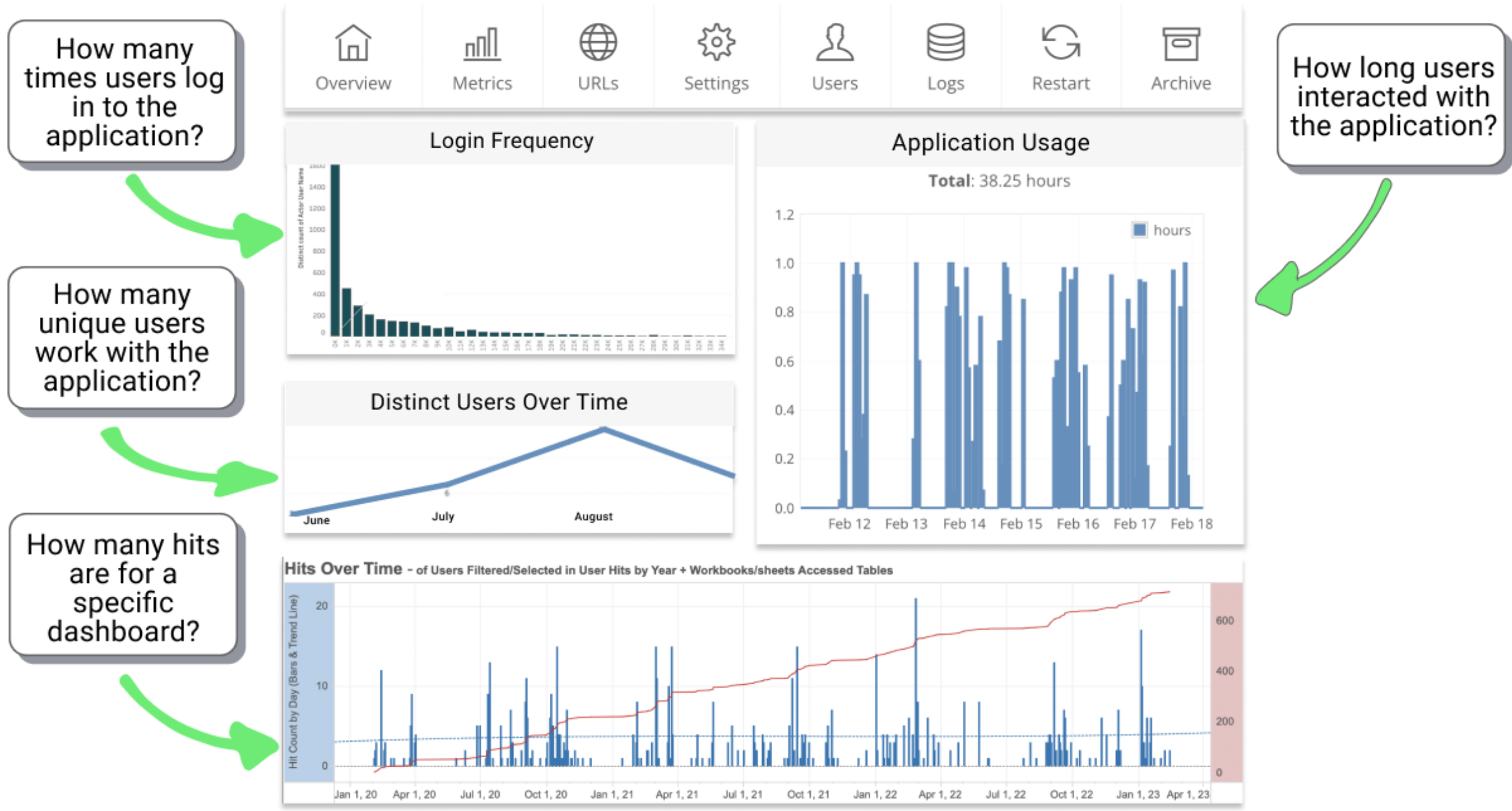
How many unique users work with the application?



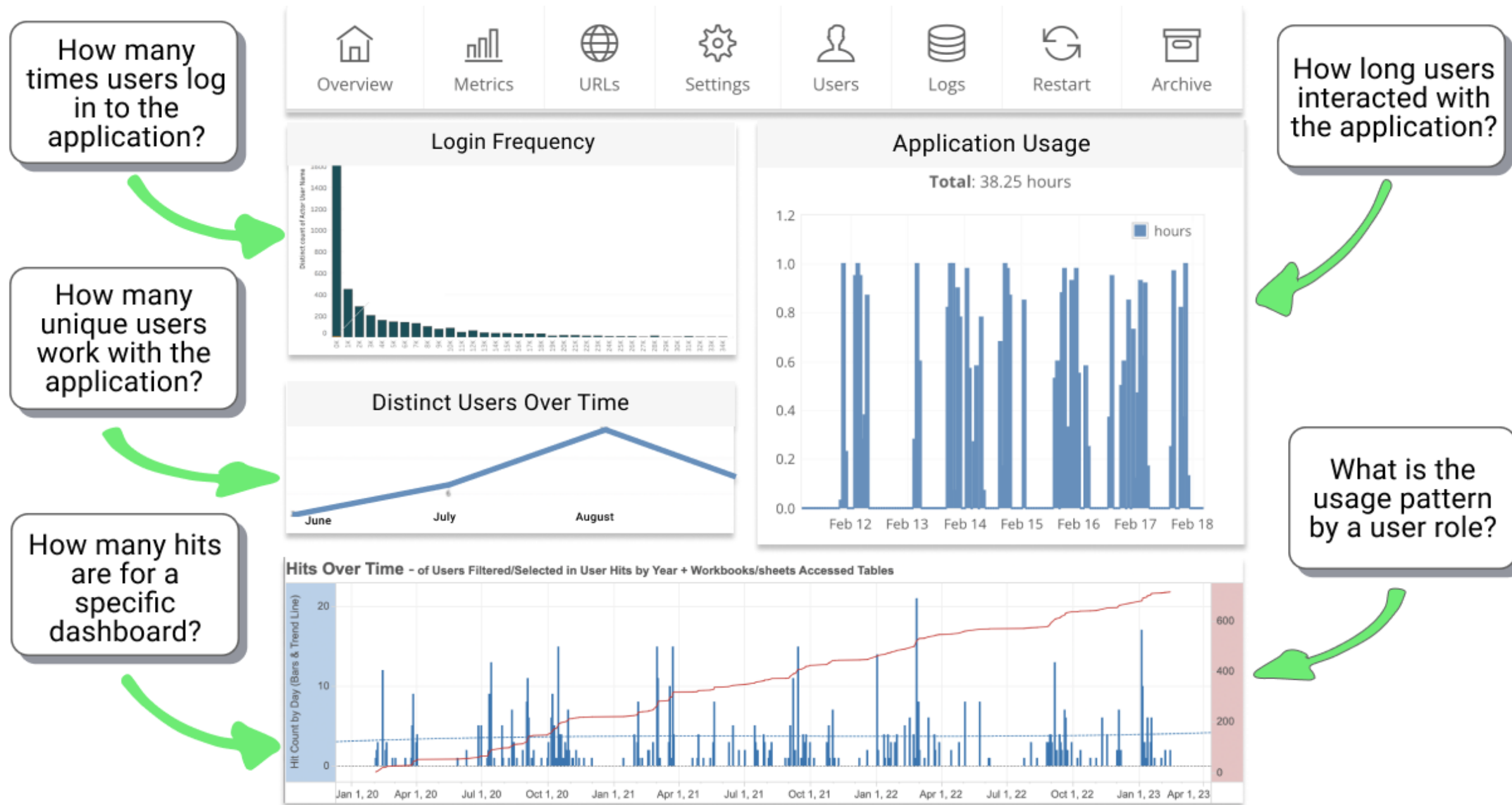
Usage analytics



Usage analytics



Usage analytics



User satisfaction

Net promoter score (NPS): how many individuals are likely to recommend your product.

How likely are you to recommend [company] to a friend or colleague on a scale of 0-10?

0 1 2 3 4 5 6 7 8 9 10

Customer satisfaction score (CSAT): how happy is the user or customer.

How would you rate your experience?

Great Okay Not Good

Business impact

Revenue impact

The revenue generated as a result of the decisions

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Time-to-action

The time it takes to make decisions and take action

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Cost savings

The cost savings achieved as a result of the decisions

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The revenue generated as a result of the decisions

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Adoption rate

How well the dashboard is being adopted and used by the intended audience

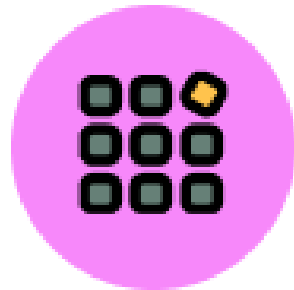
Usability testing

Hypothesis checking: testing the assumptions underlying the design of the dashboard

¹ Nielsen Norman Group. (2019). Usability Testing 101

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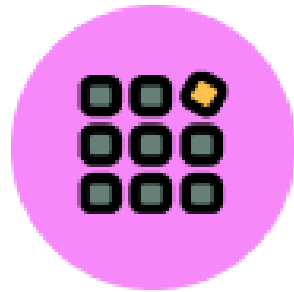


Identify problems
in the design

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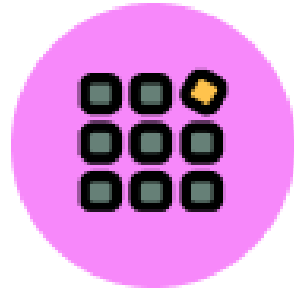


Uncover opportunities
to improve

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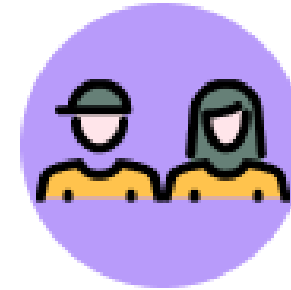
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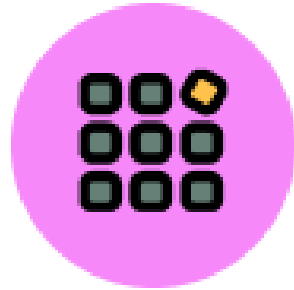


Learn about users'
behavior and preferences

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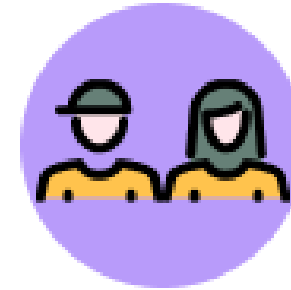
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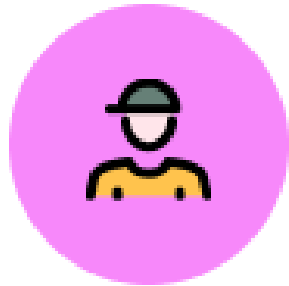


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Learn about users'
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Facilitator

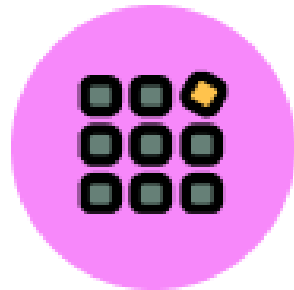


Administering tasks

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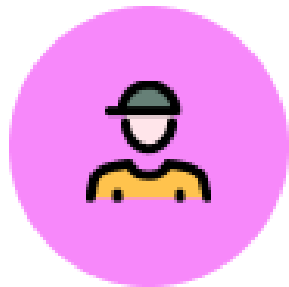
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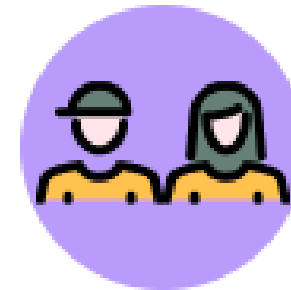


Uncover opportunities
to improve

Task



Realistic activities

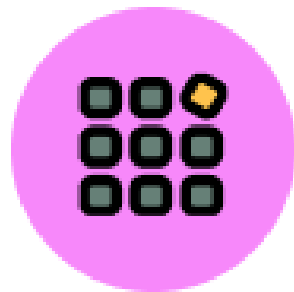


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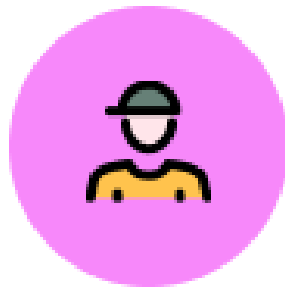
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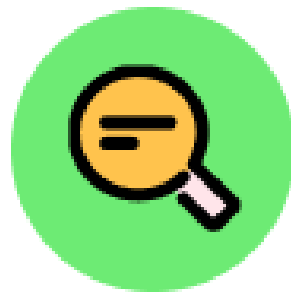


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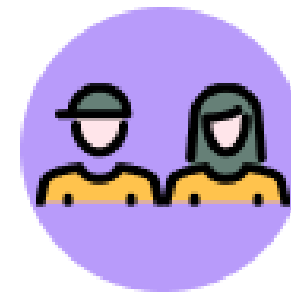


Uncover opportunities
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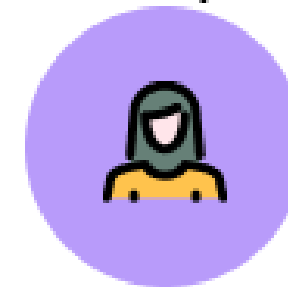


Realistic activities



Learn about users'
behavior and preferences

Participant



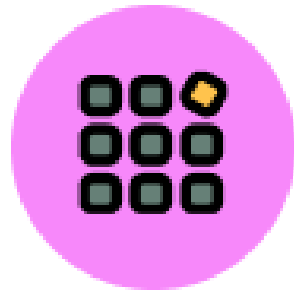
Realistic user

Qualitative testing: user experience

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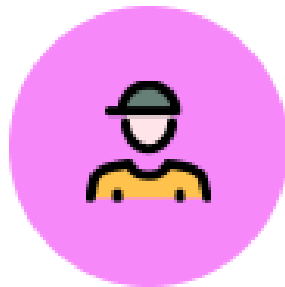
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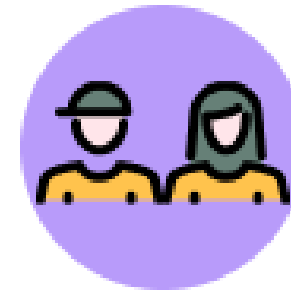


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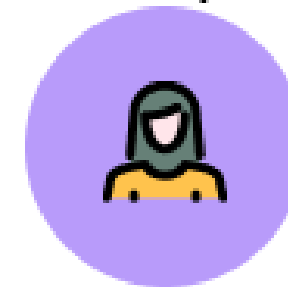


Realistic activities



Learn about users'
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Participant



Realistic user

Qualitative testing: user experience

Quantitative testing: task success, time on task

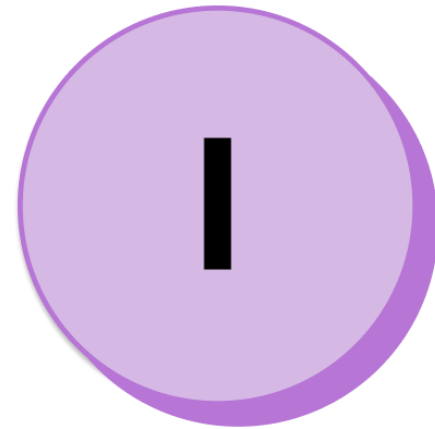
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Dashboard VITAL success



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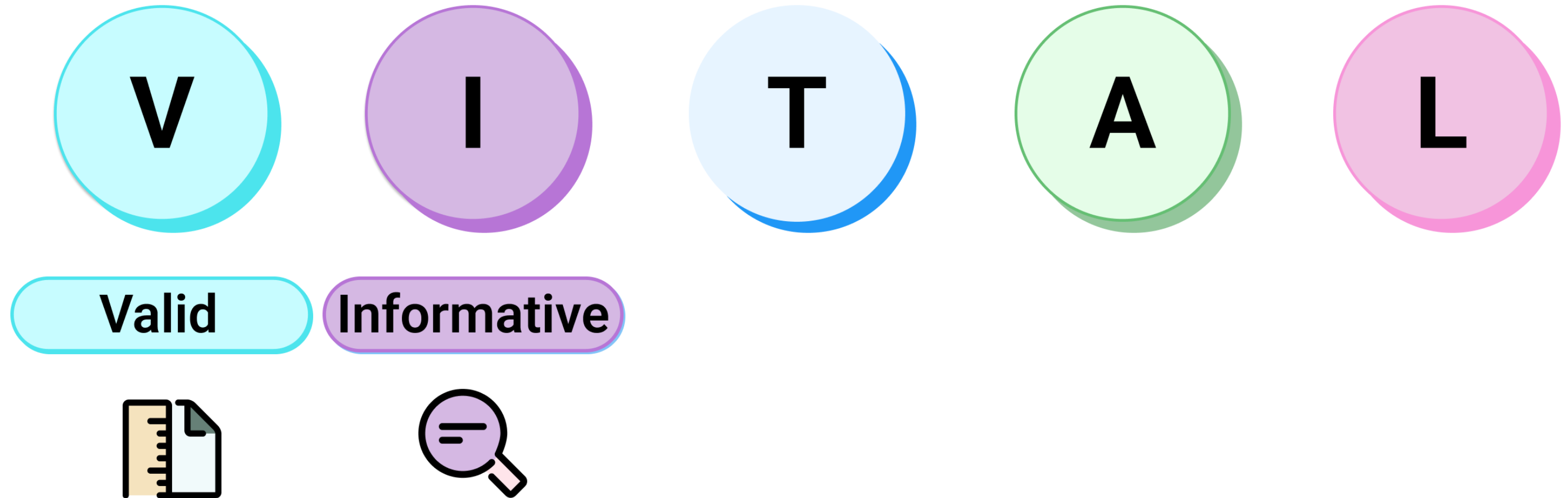


Valid



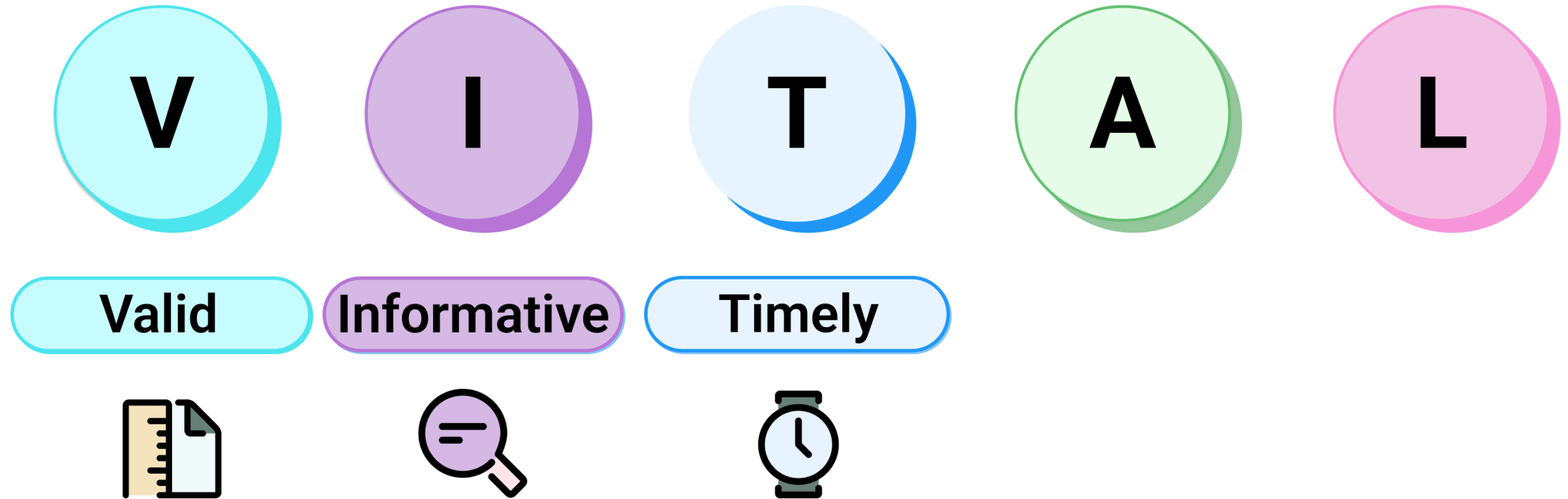
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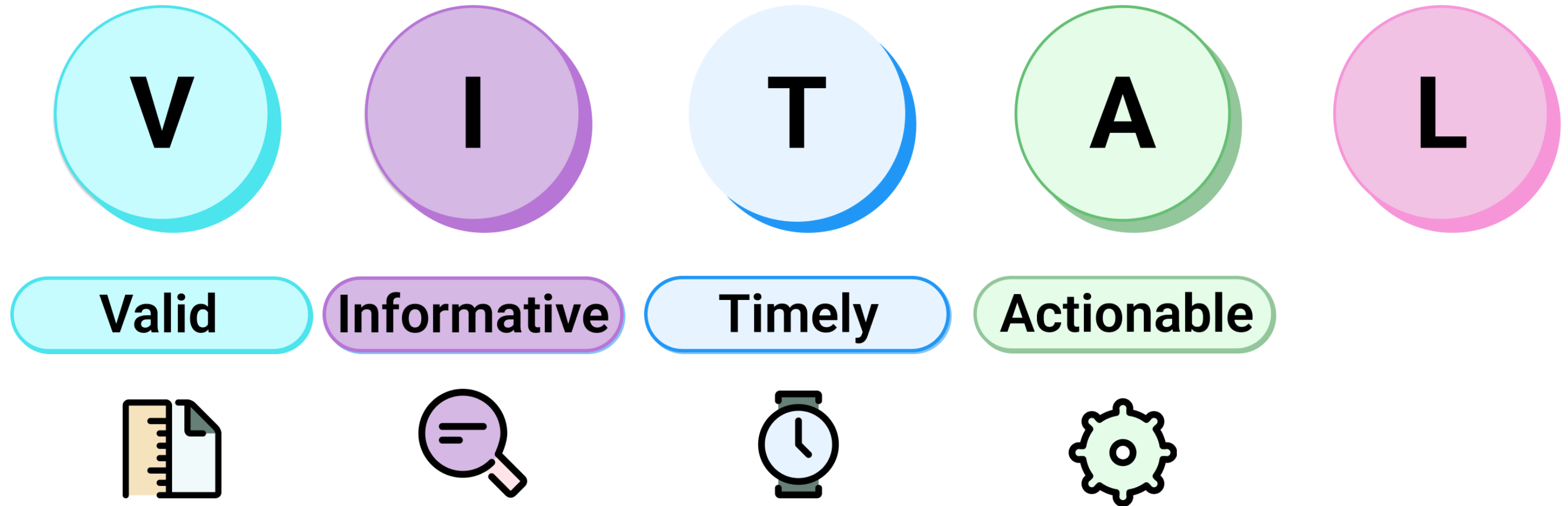
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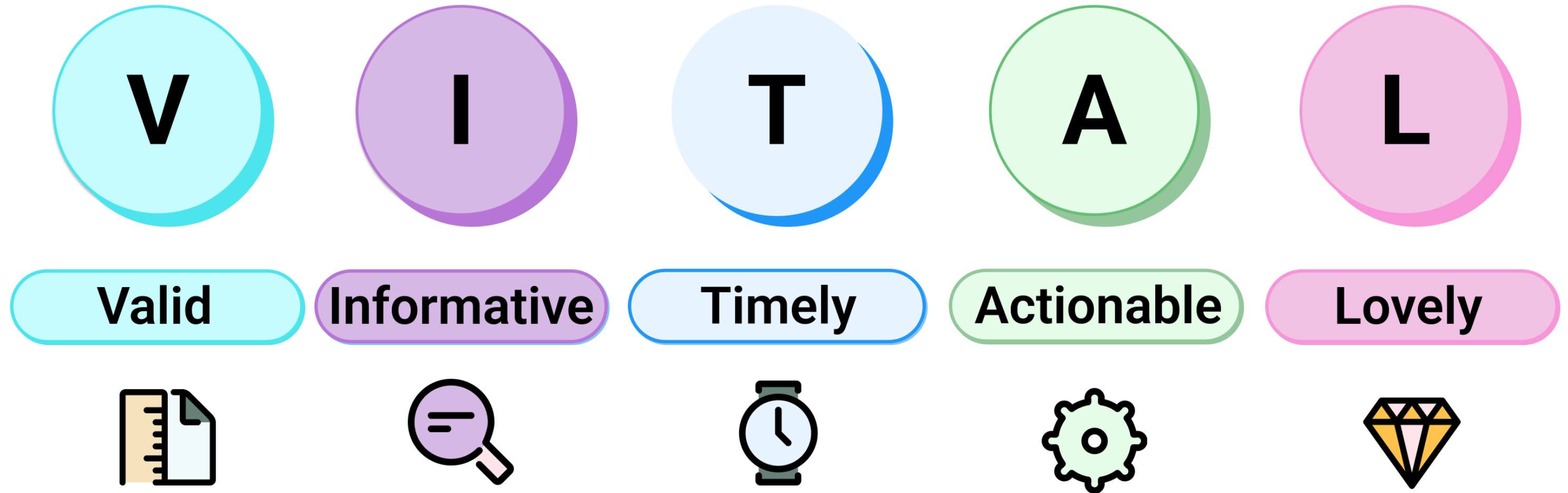
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A dashboard is only useful if people are using it

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Let's practice

DASHBOARD DESIGN CONCEPTS

Dashboard integration

DASHBOARD DESIGN CONCEPTS



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Dashboard embedding

The process of **integrating** a dashboard into a **website** or another **application**.

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Displaying insights within the context of
the application or website

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Displaying insights within the context of the application or website



Simplifying data access

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Enhancing the user experience

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Displaying insights within the context of the application or website



Simplifying data access



Enhancing the user experience



Improving communication with stakeholders

Dashboard sharing and collaboration

The process of allowing multiple stakeholders to **access**, **view**, and **interact** with a dashboard

Dashboard sharing and collaboration

The process of allowing multiple stakeholders to **access**, **view**, and **interact** with a dashboard

Single
dashboard

Multiple stakeholders: view and interact

Dashboard sharing and collaboration

The process of allowing multiple stakeholders to **access**, **view**, and **interact** with a dashboard

Single
dashboard

Multiple stakeholders: view and interact

Datasets

Stakeholders: understanding insights

Dashboard sharing and collaboration

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Single
dashboard

Multiple stakeholders: view and interact

Datasets

Stakeholders: understanding insights

Editing

Multiple stakeholders: real-time

Dashboard sharing and collaboration

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Single
dashboard

Multiple stakeholders: view and interact

Datasets

Stakeholders: understanding insights

Editing

Multiple stakeholders: real-time

Custom
access

Protecting sensitive information

Collaboration on the fly with ad-hoc dashboards

A dashboard created **on-the-fly** to address a **specific question** or provide **specific insights**

Unanticipated
Questions

Unforeseen scenarios or ideas

Collaboration on the fly with ad-hoc dashboards

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Data exploration

Ability to explore different pathways

Collaboration on the fly with ad-hoc dashboards

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Unanticipated Questions

Unforeseen scenarios or ideas

Data exploration

Ability to explore different pathways

Freedom

Overcoming a fixed and rigid perspective

Dashboard integration using entrepreneurial mindset

An approach for embracing creativity, innovation, and changes

Dashboard integration using entrepreneurial mindset

An approach for embracing **creativity**, **innovation**, and **changes**

Solving business
problem

Creating dashboards for solving real-
world problems

Dashboard integration using entrepreneurial mindset

An approach for embracing **creativity**, **innovation**, and **changes**

**Solving business
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Creating dashboards for solving real-
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**Embracing
experimentation**

Experimenting with new visualization
techniques

Dashboard integration using entrepreneurial mindset

An approach for embracing **creativity**, **innovation**, and **changes**

**Solving business
problem**

Creating dashboards for solving real-world problems

**Embracing
experimentation**

Experimenting with new visualization techniques

Leveraging data

Using data to drive insights

Dashboard integration using entrepreneurial mindset

An approach for embracing **creativity**, **innovation**, and **changes**

Solving business problem

Creating dashboards for solving real-world problems

Embracing experimentation

Experimenting with new visualization techniques

Leveraging data

Using data to drive insights

Iterating and improving

Continuously evolving to meet the needs

Using dashboards in **innovative** and **effective ways** to drive **business success**

Let's practice!
DASHBOARD DESIGN CONCEPTS

Congratulations!

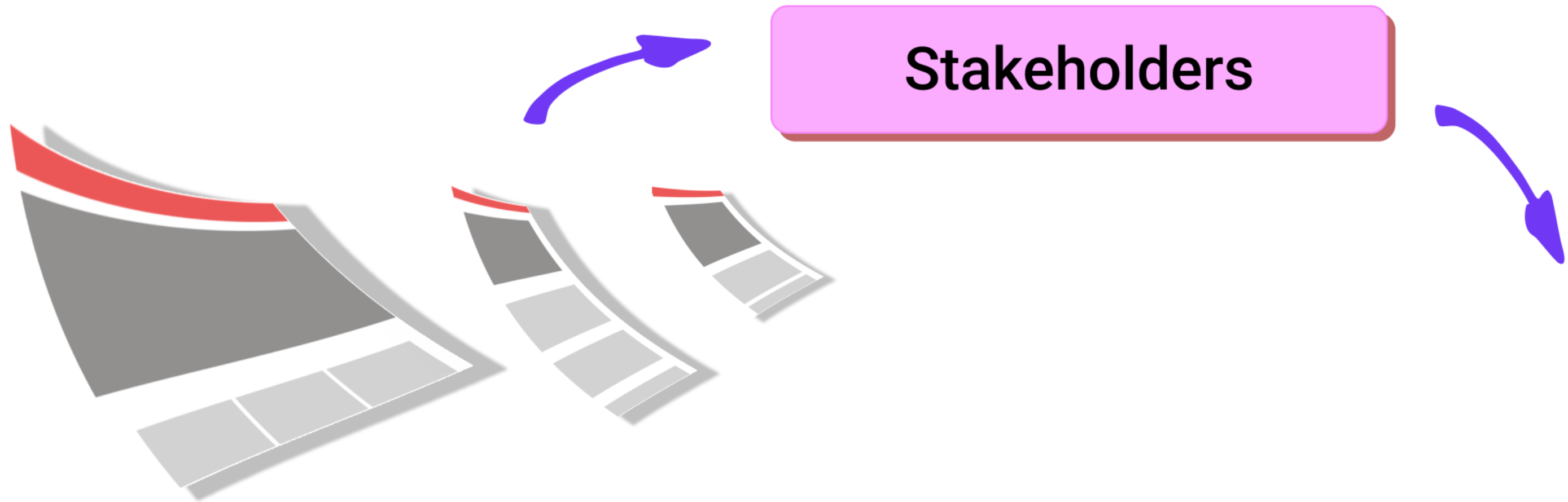
DASHBOARD DESIGN CONCEPTS



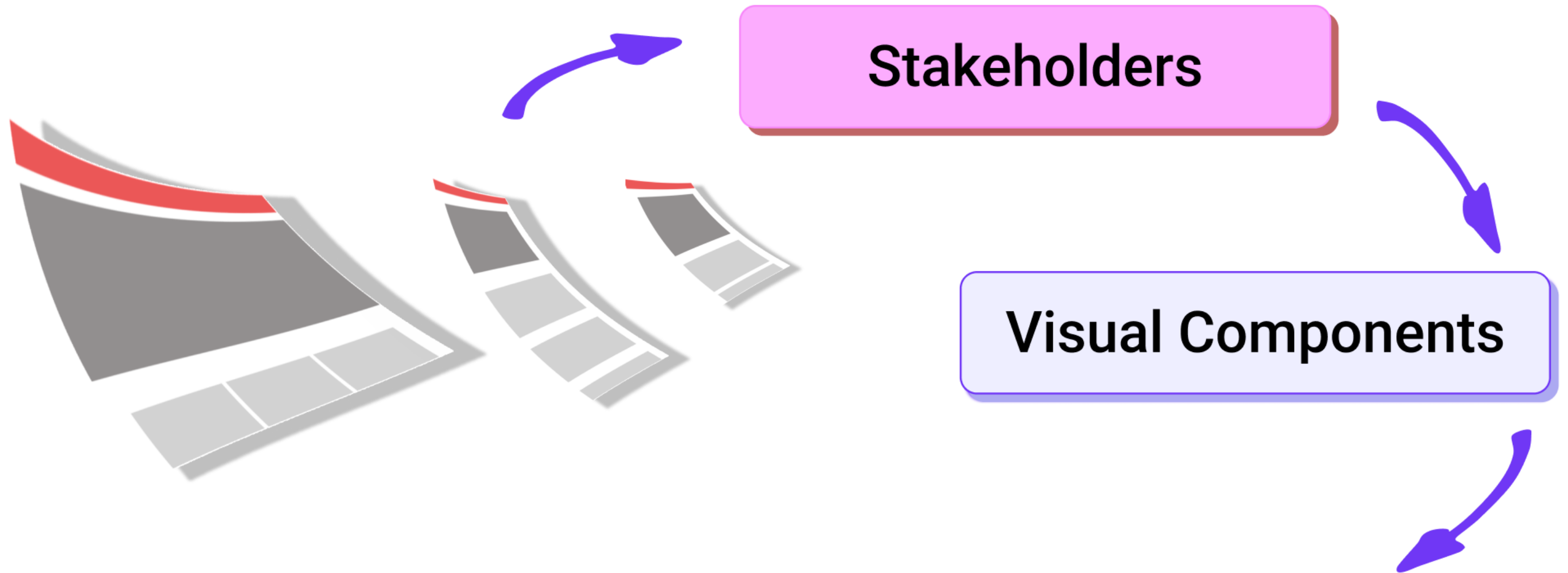
Olga Scrivner

President, Scrivner Solutions Inc

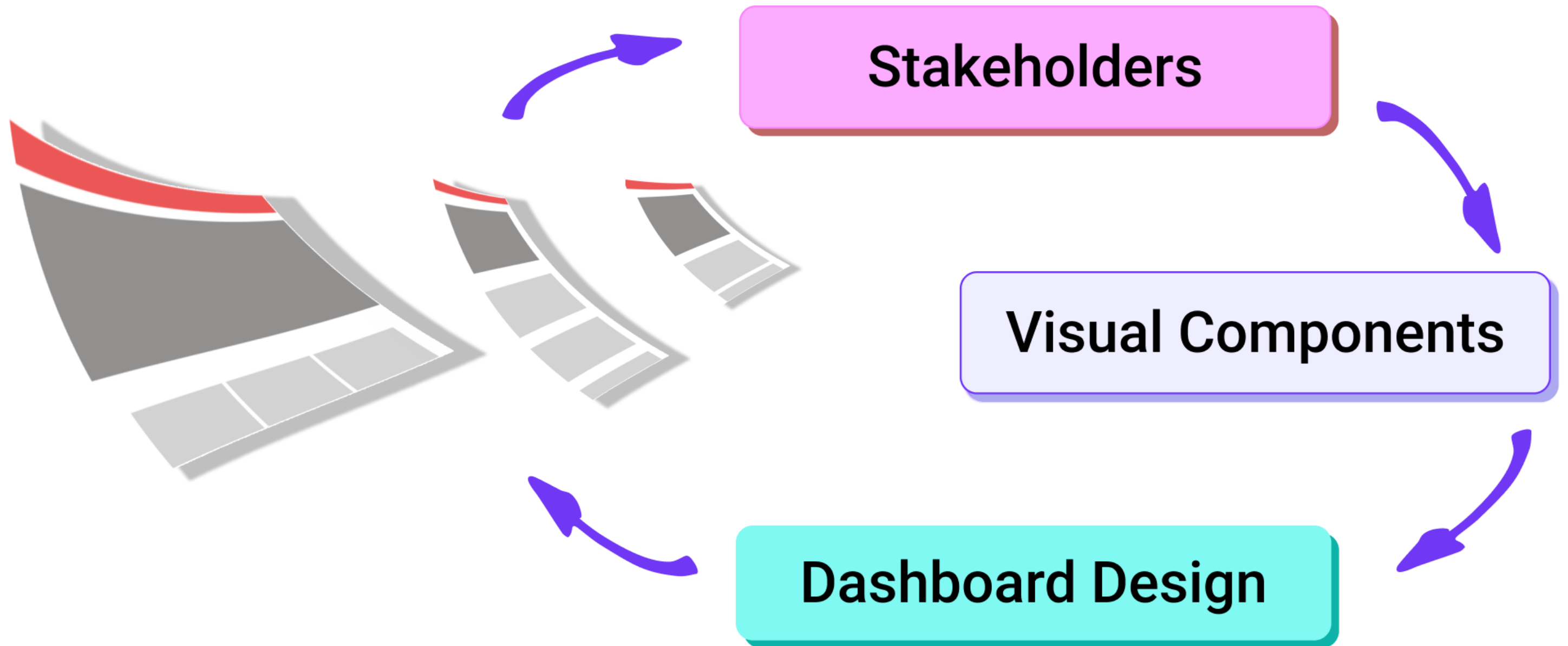
What you've learned



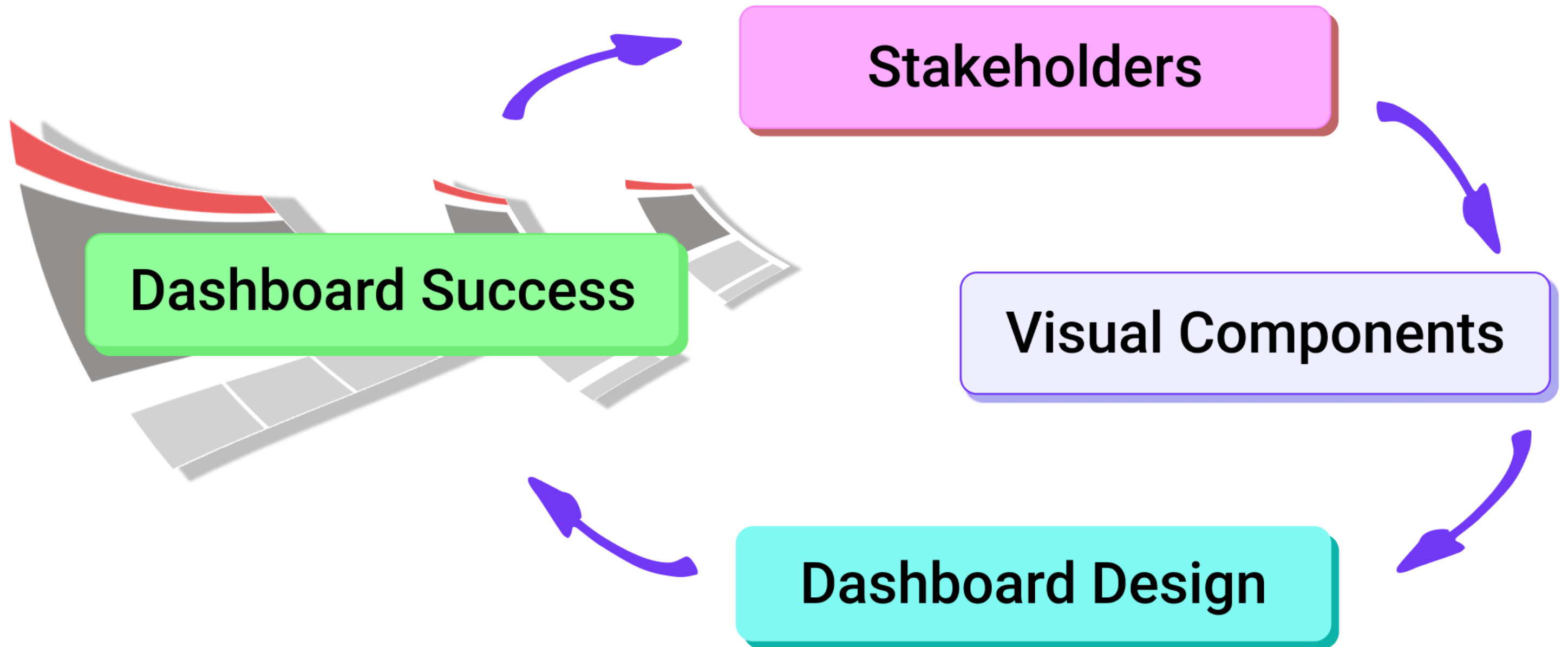
What you've learned



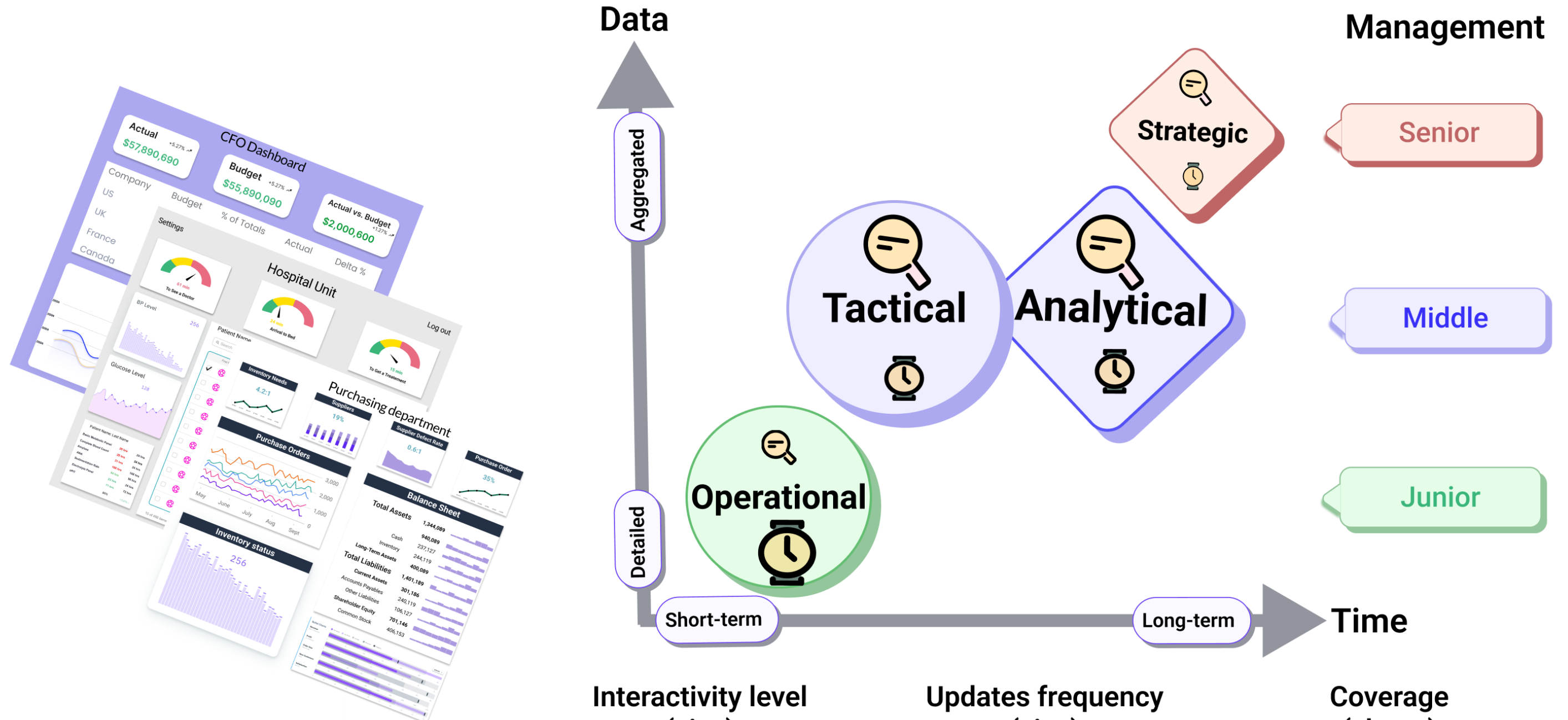
What you've learned



What you've learned



What you've learned



What you've learned



Congratulations!

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